Applied AI & Machine Learning Capstone Project

Course: Data Science for Managers | Executive Assignment Specification

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1. Executive Overview & Objectives

In teams of 8 to 9 students, you will design, build, and evaluate an end-to-end, AI/ML-driven business product. The primary goal of this capstone assignment is to bridge technical machine learning capabilities with executive decision-making and tangible business ROI.

This course encourages modern 'Vibe Coding' workflows—leveraging state-of-the-art AI orchestration tools (such as VS Code, GitHub Copilot/Codex, Google Antigravity, Google AI studio, or Claude Cowork) to accelerate user interface construction, boilerplate setup, and feature engineering. However, your team will be rigorously evaluated on technical depth and theoretical understanding. You must be able to explain all underlying algorithms, and architecture choices without relying on black-box assumptions.

> **THE GOLDEN RULE OF VIBE CODING:** *AI agents can generate your interface and code scaffolding, but YOU must own the technical stuff and algorithm behavior. Your team must be able to explain the underlying model, parameter selection, work flow, choices etc.*

2. Key Project Deliverables

A. Interactive Business Product (Working Prototype)

- An intuitive web application built in Python (may be using Streamlit, Gradio, or Shiny for Python). Non-technical executives must be able to interact with your model seamlessly.
- A production-grade Python ML pipeline (scikit-learn, XGBoost, PyTorch, or Hugging Face transformers) executing real-time inferences.

B. Technical & Business Presentation (15 Minutes + Q&A)

- The Executive Pitch (5 Minutes): Business opportunity, target persona, financial ROI / business metrics, and market strategy.
- The Technical Deep Dive (10 Minutes): System architecture diagram, feature selection logic, model optimization, cross-validation results, and error analysis.
- Live Interactive Demonstration: Performing live inference using realistic test data directly on your application interface.

C. Code Repository & Development Audit Log

- A clean, modular GitHub repository structured into src/data.py, src/model.py, and app.py with a complete requirements.txt.
- A 1-page Vibe Coding Log detailing prompt engineering strategies, agentic workflows, and platforms utilized (e.g., Antigravity, Claude Cowork, Codex).

3. Sample Project Tracks & Business Domains

Track / Focus Area	Required AI/ML Algorithms	Example Business Product
1. Customer Intelligence	K-Means, DBSCAN, Hierarchical Clustering, RFM Analysis	SegmentIQ: Customer Churn & Lifetime Value Optimizer with automated campaign targeting.
2. Decision Automation	Logistic Regression, XGBoost, Random Forest, SHAP	CreditRisk AI: Real-time commercial loan underwriting engine with regulatory compliance audit trails.
3. Recommendation Systems	Collaborative Filtering (SVD/Matrix Factorization), Content-Based	RetailBoost: B2B cross-sell and up-sell engine with dynamic price-elasticity adjustments.
4. Predictive Operations	Ridge/Lasso Regression, ARIMA/Prophet, Neural Networks	DemandPulse: Supply chain inventory forecaster minimizing warehouse holding costs.
5. Enterprise GenAI / NLP	Fine-tuned LLMs, RAG (Vector DBs), Distance Metrics	ComplianceGuard: Enterprise contract risk scanner extracting liability and indemnity clauses.

4. Evaluation & Grading Rubric (100 Points Total)

Evaluation Criteria	Weight	Key Assessment Questions
1. Business Value & Product Usability	**30 Points**	Is the problem well-scoped? Is ROI explicitly quantified? Can a non-technical executive seamlessly use the interactive web app?
2. Technical Rigor & Pipeline Design	**40 Points**	Are preprocessing, scaling, hyperparameter tuning, and cross-validation executed properly? Is model explainability (SHAP/LIME) implemented?
3. 'Under the Hood' Technical Defense	**20 Points**	Can every team member explain the underlying technical stuff, and parameter choices during individual Q&A?
4. Engineering Efficiency & Vibe Coding	**10 Points**	How effectively were AI agents (Codex, Antigravity, Claude Cowork) orchestrated? Is the GitHub repository clean, modular, and reproducible?